Track 2 report — which approved medicines could act on this child's broken checkpoint?

Standing: draft, written 2026-09-18 from executed runs (iteration 2, step 7 — methodology entries 110–113). Every number is read from a table with a provenance manifest, every claim was registered before its test, and every cited paper was resolved against PubMed the same day. **Not uploaded.** Track 2 accepts one submission; this folder will be re-read and re-rendered before it goes.

Written for the panel, in plain words. Each technical term appears once with its meaning in brackets; the full dictionary is served at `/learn` on the dashboard.

> The challenge's own framing, kept in view throughout: *submissions are hypotheses for follow-up, not evidence that a medicine works.* Nothing below is a recommendation to treat.

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1. The answer in one page

The child. A boy with mosaic variegated aneuploidy (MVA: in some of his cells, a chromosome is missing or duplicated, and different cells are wrong in different ways), short stature, failure to thrive, calcium deposits in the kidneys, and a rhabdomyosarcoma (a muscle-tissue cancer of childhood). Eight clinical terms were delivered with the case.

The cause. Track 1 found, and the challenge's answer key confirmed, two different broken copies of one gene, `BUB1B` — one from each parent (a *compound heterozygote*). The gene makes BubR1, the protein that acts as the **brake** on cell division: it holds a dividing cell until every chromosome is properly attached before the cell is allowed to split. One copy makes no protein at all (a *stop* change); the other makes a weakened protein (a *missense* change in the protein's last stretch). The child therefore has **some brake, but less than normal** — which is exactly the pattern every published MVA case with this gene shows, because having no brake at all is not survivable.

The mechanism, in the form the methods form asks for. *Loss of function*, not gain: less BubR1, a leakier checkpoint, chromosomes that separate too early, cells with the wrong count, and — in surviving cells — a chronic stress from carrying the wrong number of copies of many proteins. Downstream: the aneuploidy that names the disease, the growth problems, and a predisposition to exactly the cancers this child has.

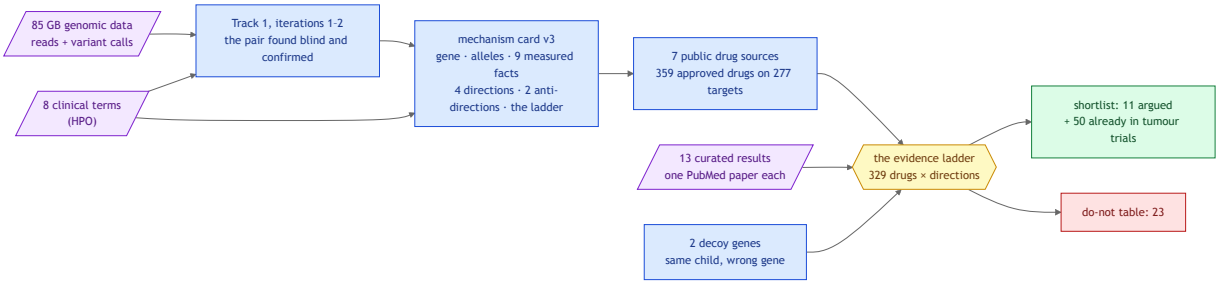
Four ways a drug could act, graded on a ladder written before the search.

Direction	The idea	What we found	Best rung
D1 reinforce the brake	make the remaining checkpoint stronger	**no approved drug does this.** Every checkpoint drug on the market or in trials *weakens* it. D1 became the do-not table	—
D2 exploit the stress of aneuploid cells	aneuploid cells are already stressed; some approved drugs tip them over	**chloroquine** (a malaria drug) killed aneuploid cells preferentially in the reference screen; metformin, hydroxychloroquine and bortezomib inherit the result through their drug class	2 (cell model)
D3 raise the amount of working BubR1	the weakened copy still works; stabilise it	**niacin / nicotinamide** (vitamin B3 forms): boosting the NAD^+ pathway raised BubR1 in mice bred to have too little of it; valproic acid and vorinostat act on BubR1's acetylation switch	3 (animal model)
D4 the child's actual tumour	approved, non-chemotherapy drugs with a mechanism in rhabdomyosarcoma	**statins** (lovastatin, simvastatin, atorvastatin): rhabdomyosarcoma cells depend on making their own cholesterol; lovastatin itself slowed them; simvastatin has already been in a completed paediatric sarcoma trial	5 (approved, trialled in this tumour, used in children) — carried by rung-2 mechanism evidence

The half a clinician may care about most: what not to do. Thirteen approved drugs are *spindle poisons* — they work by jamming cell division and relying on the checkpoint to hold the cell. In a child whose checkpoint is the broken part, that reliance is a hypothesis worth weighing before treatment. Several of them (vincristine, vinorelbine) are standard rhabdomyosarcoma chemotherapy. **This report flags the question for the treating team; it does not advise withholding standard care.**

What we do not know. How aneuploid this child's cells actually are (the sequencing data delivered cannot measure it with any power; the analysis that could is planned and not run). Whether the weakened BubR1 copy is unstable or simply inactive. And whether any drug on this page helps — rung 3 is one mouse paper.

2. Where this comes from — two datasets, one line of evidence



> **Legend** — **purple = raw inputs** (the child's DNA, the child's clinical terms, and the papers a person read); **blue = machinery** (every box is a tool with a manifest); **yellow = the decision point** (the ladder); **green = the goal** (the shortlist); **red = what to avoid** (the do-not table).

Both datasets are used at every stage: the genomic data fixed the gene and the two alleles; the clinical terms fixed the disease, the tumour and — through the tumour — the fourth direction and the paediatric questions. Where a result uses only one, the table says so.

3. The mechanism (methods form Q14: loss- versus gain-of-function, pathway, consequence)

3.1 Loss of function, of a specific shape

copy	what changed	what it does to the protein	evidence
one	a *stop* change in exon 17 of 23	the protein ends early, before its last working part — **no functional protein**	catalogued as disease-causing for MVA (ClinVar VCV000533901)
the other	a *missense* change in exon 23 of 23 — one amino acid swapped	a **weakened** protein: never seen in 140,000 healthy genomes, predicted damaging by two independent predictors	Track 1 (iteration 1, entries 070–074; iteration 2, entry 099)

Nine facts measured in iteration 2 sit on the mechanism card so the argument does not rest on the allele labels alone:

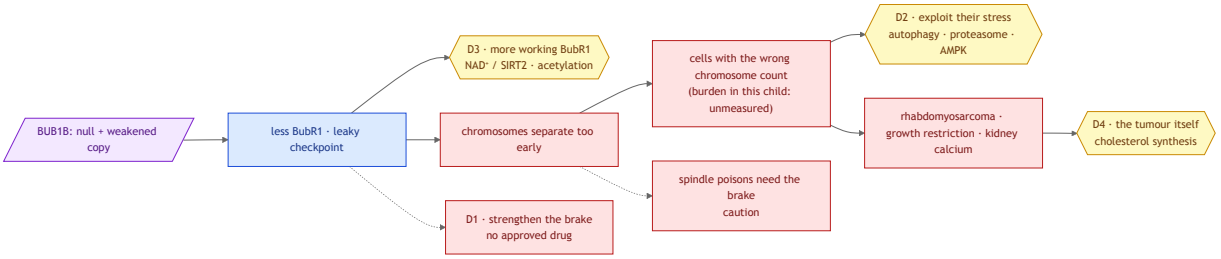
fact	measured	what it changes for a drug
the two copies are on **different** chromosomes (*trans*)	**confirmed** by the challenge's answer key (a full match at rank 1)	the whole argument stands on a confirmed genotype, not an inference
each copy present **once** in the raw sequencing reads	yes — both alleles at one-copy dosage, read without an aligner	a *null + weakened* genotype: some BubR1 exists to stabilise (D3 is live)
no deleted or duplicated stretch in the gene	every exon at two copies; no structural variant in the gene	the two alleles are the whole story
where the missense sits in the protein	outside every catalogued domain, in a region the structure model is confident about	we **cannot** say whether the weakened copy is unstable or inactive; D3 is argued as *more BubR1*, not *refold it*
how aneuploid this child's cells are	**unmeasured** — the karyotype test iteration 1 ran has no power for mosaic gains; the powered analysis (step 8) has not run	D2 is argued from published aneuploid-cell biology, not from this child's burden

(Complete list F1–F9, with the entry each comes from: mechanism card v3, and step 7's README.)

3.2 The pathway: the spindle assembly checkpoint

BubR1 is a core part of the **spindle assembly checkpoint** (SAC — the wait-until-attached brake) and of the complex that switches off the machinery that pulls chromosomes apart. Less BubR1 means a **leaky brake**: division can proceed with chromosomes still unattached. Published MVA patient cells show strongly reduced BubR1 and a weakened checkpoint (Suijkerbuijk 2010, PMID 20516114); the founding paper tied biallelic `BUB1B` changes to MVA with cancer predisposition (Hanks 2004, PMID 15475955). Complete loss is lethal before birth in mouse and human — which is why "switch BubR1 off" is never a therapeutic direction here.

3.3 Downstream consequences, and where a drug could stand



> **Legend** — **purple** the genotype; **blue** the molecular consequence; **yellow** the three places an approved drug could stand, each a direction below; **red** what goes wrong, and the two dotted boxes are the direction with no drug and the class to be cautious with.

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4. How the candidates were found and graded

4.1 The search: one card, seven public sources, no hand-typed query

The mechanism card (a structured file naming the gene, the two allele classes, the pathway, the disease, the tumour and four *directions* with seed genes) drives every query. The Reactome pathway map widens the seed genes into 277 targets; Open Targets, ChEMBL and DGIdb list every drug known to act on them (3,859 drug–target rows, **359 market-approved drugs**); openFDA confirms a current US label; ClinicalTrials.gov and PubMed answer the trial and literature questions. Every table carries a manifest with the retrieval time and a digest of every reply, and the same command re-runs it for any other case by swapping the card.

4.2 The ladder: written before the search

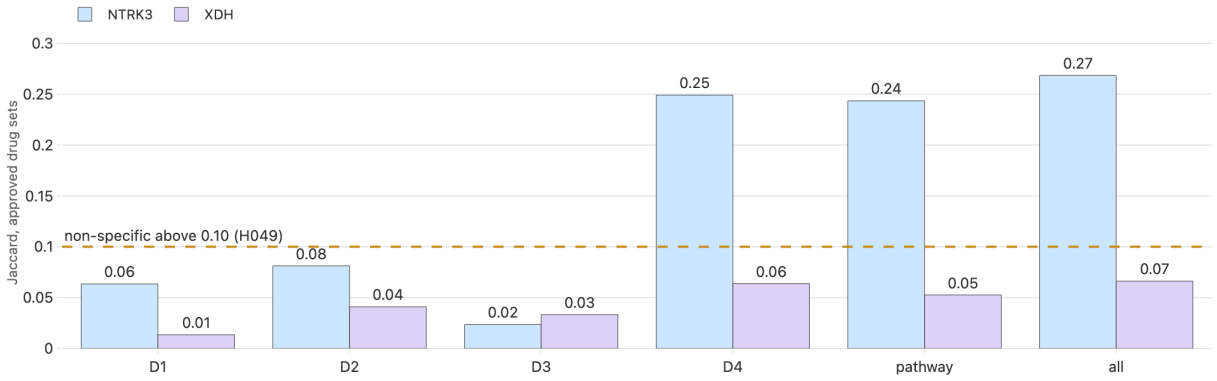
Every approved drug, on every direction it touches, is placed on **the highest rung it passes**:

rung	meaning	how it is tested
5	approved for a related indication, with paediatric use	approved **and** a registered trial in the child's tumour **and** a paediatric pointer (a label section 8.4, or a registered study admitting children)
4	clinical trial in a related indication	a registered trial in the child's tumour
3	animal model of the mechanism	a paper, read by a person, with its PubMed id
2	cell model of the mechanism	a paper, read by a person, with its PubMed id
1	mechanism only	the drug acts on a target in the direction's set — *named, never submitted*

Two rules the ladder prints rather than hides. **Automated and curated tests sit side by side:** rungs 4–5 are looked up by a program and say "already being tried in this tumour"; rungs 2–3 come from a paper a person read, and the row carries the PubMed id. **Inherited evidence is marked:** when the paper tested a same-axis compound and not the approved drug (the screen used AICAR; the approved drug is metformin), the row says `proxy` and names what was tested.

4.3 A control the search had to pass: the same child, a different gene

Track 1 tested every ranker against decoy genes; Track 2 does the same. The mechanism card was copied with the child's clinical side kept and the gene swapped — once for **NTRK3**, drawn by a seeded rule from Track 1's matched decoy set, once for **XDH**, a metabolic enzyme with nothing to do with cell division — and the unchanged pipeline ran on both. Whatever came out for the decoy as readily as for `BUB1B` was nominated by the clinical queries or by database density, not by the mechanism.

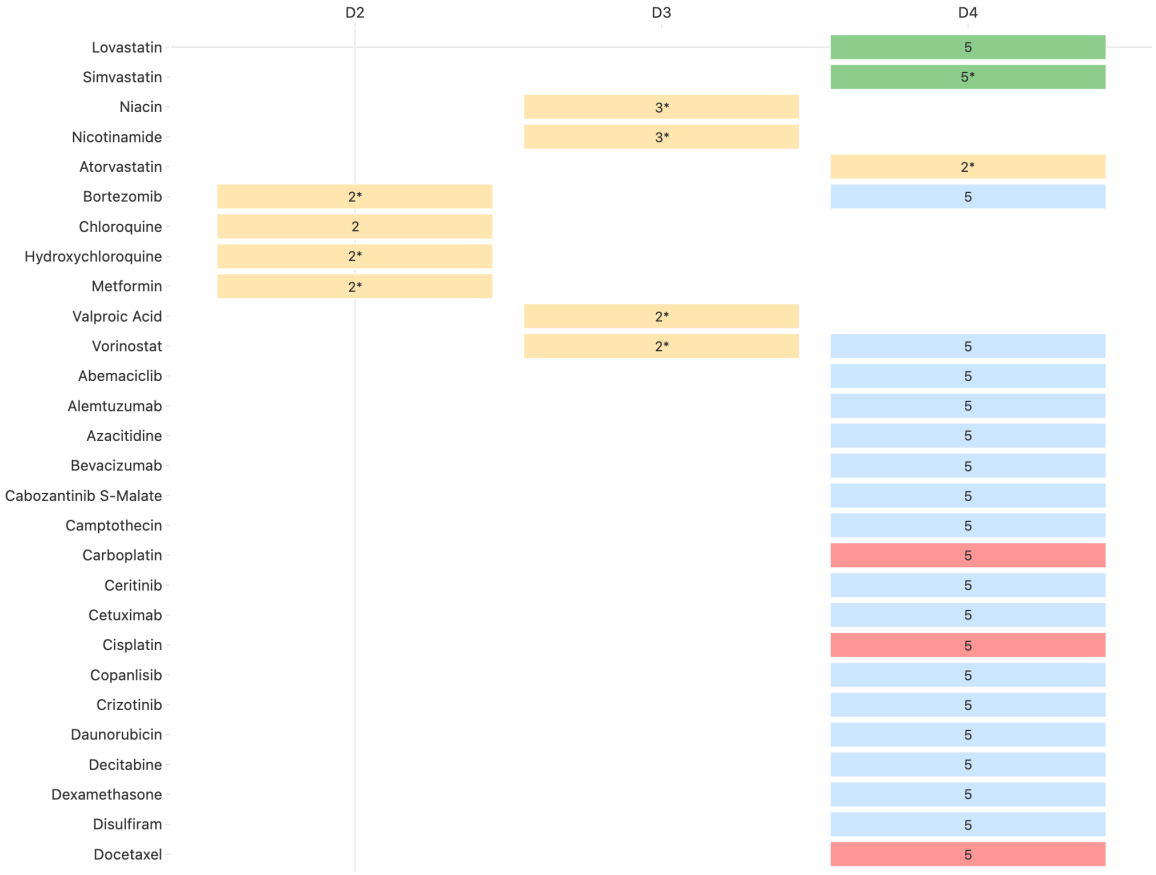


> **Legend** — one bar per decoy gene; the dashed yellow line is the pre-registered 0.10 threshold. **D4 (the tumour direction) is above it** against NTRK3 (Jaccard 0.25): the cancer pharmacopoeia comes out for any cancer-adjacent gene. **D1–D3 are below it** against both decoys: the mechanism directions are specific to the gene. The consequence, applied: D4 is argued only from curated mechanism results, never from "this drug has a rhabdomyosarcoma trial".

The control also caught a flaw in the first grading run: a rhabdomyosarcoma trial was lifting drugs onto rung 5 on *mechanism* directions (85 drugs, tetracycline among them). A trial in the tumour says nothing about a checkpoint; the rule was narrowed to the tumour direction and the run repeated before any verdict — the change is recorded with before/after counts in entry 113.

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5. The candidates



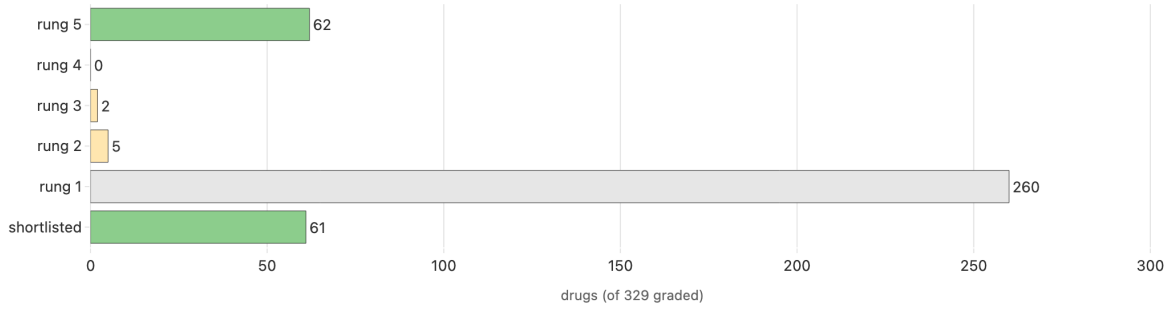
> **Legend** — the number is the rung; **green** a curated result and rung 4–5; **yellow** a curated result at rung 2–3; **blue** the automated tests alone (already being tried in the tumour — not a finding of this work); **red** set aside (a spindle poison or the standard chemotherapy); `*` inherited (proxy) evidence. Drugs with a curated result are listed first.

5.1 The eleven drugs with a cited mechanism result

#	drug (approved for)	direction	rung	the cited result	inherited?	given to children?
1	**lovastatin** (cholesterol)	D4 tumour	5	lovastatin itself slowed rhabdomyosarcoma cells; the tumour depends on making its own cholesterol and cannot take it up instead (Gizaw 2025, PMID 41249736). A completed phase-1 study gave a statin with chemotherapy to children with relapsed solid tumours including rhabdomyosarcoma (NCT02390843)	no	label section 8.4; studies admitting children
2	**simvastatin** (cholesterol)	D4	5	same class result; it is the statin in NCT02390843	yes (lovastatin)	studies admitting children
3	**nicotinamide** (vitamin B3)	D3 more BubR1	3	the NAD^+ precursor NMN raised BubR1 in mice, and more SIRT2 (the enzyme that stabilises BubR1) extended the life of BubR1-deficient mice (North 2014, PMID 24825348)	yes (NMN)	studies admitting children
4	**niacin** (cholesterol; vitamin B3)	D3	3	same result; niacin is the prescription-strength NAD^+ precursor	yes (NMN)	studies admitting children
5	**chloroquine** (malaria)	D2 stress	2	chloroquine itself preferentially killed cells with an extra chromosome in the reference screen (Tang & Amon 2011, PMID 21315436)	**no**	studies admitting children
6	**bortezomib** (myeloma)	D2	2	aneuploid human cells depend on degrading proteins faster and are more sensitive to proteasome inhibitors; myeloma patients' aneuploidy tracked their response (Ippolito 2024, PMID 39247952; Chui 2019, PMID 31530568)	yes (the drug class)	label 8.4; studies admitting children
7	**valproic acid** (epilepsy)	D3	2	HDAC2/3 strip the acetyl mark that protects BubR1 from being degraded (Park 2017, PMID 28985013; Choi 2009, PMID 19407811); an HDAC inhibitor would be expected to slow that loss — inferred, not shown	yes (class mechanism)	label 8.4; studies admitting children
8	**metformin** (diabetes)	D2	2	the AMPK activator AICAR was aneuploidy-selective in the same screen; metformin is the approved AMPK activator	yes (AICAR)	label 8.4; studies admitting children
9	**vorinostat** (lymphoma)	D3	2	same class mechanism as valproic acid	yes	label 8.4; studies admitting children
10	**hydroxychloroquine** (lupus, arthritis)	D2	2	chloroquine's result, inherited by its clinical analogue	yes (chloroquine)	label 8.4; studies

#	drug (approved for)	direction	rung	the cited result	inherited?	given to children?
						admitting children
11	**atorvastatin** (cholesterol)	D4	2	same class result as lovastatin; carries a paediatric indication on its label	yes (lovastatin)	label 8.4; studies admitting children

Reading the table honestly: **one** drug has a molecule-level animal result on the checkpoint itself — none. The deepest mechanistic lead (D3, more BubR1) rests on one mouse paper and is inherited by two vitamin-B3 forms. The best-supported *tumour* lead (statins) has a completed paediatric trial behind it and a molecule-level cell result. The only molecule-level *aneuploidy* result is chloroquine's.



> **Legend** — drugs counted once at their best rung; **green** rung 4–5 and the shortlist, **yellow** rungs 2–3, **grey** rung 1 (named, never submitted). Most of the 62 at rung 5 are there by the automated test alone: approved, trialled in rhabdomyosarcoma, used in children.

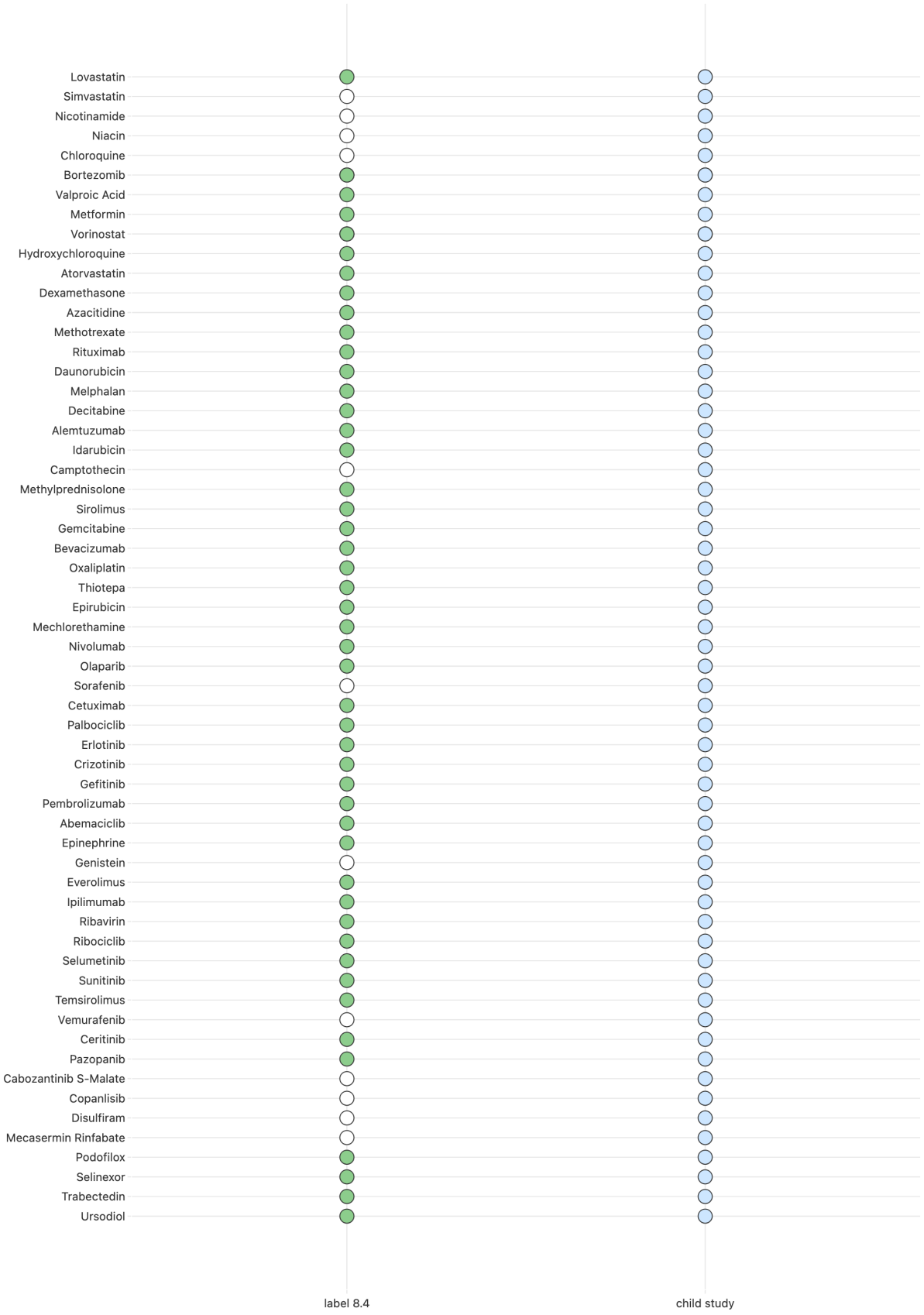
5.2 Fifty drugs already being tried in rhabdomyosarcoma — listed, not claimed

Fifty more approved drugs reach rung 5 automatically: they have a registered rhabdomyosarcoma trial and a paediatric pointer (dexamethasone, azacitidine, methotrexate, bevacizumab, the CDK4/6 and mTOR inhibitors, the checkpoint antibodies…). They are in `shortlist.tsv` with the verdict *context trial only – not a mechanism finding*, because the decoy control showed the same list comes out for a gene that has nothing to do with the checkpoint. They are the tumour's trial landscape, and a reader deserves to see them; they are not this work's finding.

5.3 The do-not table

verdict	drugs	why
caution — spindle poisons (13)	vincristine, vinorelbine, vinblastine, vinflunine, paclitaxel, docetaxel, cabazitaxel, eribulin, ixabepilone, epothilone D, colchicine, brentuximab vedotin, polatuzumab vedotin	these drugs work by jamming the division machinery and *relying on the checkpoint to hold the cell*; MVA cells have a weakened checkpoint and slip through (Suijkerbuijk 2010, PMID 20516114; Sinha 2019, PMID 30601084). Vincristine and vinorelbine are standard rhabdomyosarcoma chemotherapy: **a hypothesis for the treating team to weigh, never advice to withhold treatment**
do not — further checkpoint weakening (0 approved)	Mps1/TTK and Aurora inhibitors	aneuploid cancer cells are selectively killed by weakening the checkpoint further (Cohen-Sharir 2021, PMID 33505028) — the same logic makes a child whose checkpoint is constitutionally weak the worst possible candidate. None is approved; every one in the candidate table is phase 3 or earlier
standard of care (10)	vincristine, dactinomycin, cyclophosphamide, ifosfamide, doxorubicin, etoposide, irinotecan, temozolomide, carboplatin, cisplatin, topotecan	already the treatment for the tumour; giving them is not a repurposing. Named so a reader sees they were considered, not missed

5.4 Has each candidate been given to children?



> **Legend** — **green** a US label with a *Pediatric Use* section (8.4); **blue** at least one registered study admitting children, in any indication; empty = neither found. Every argued drug has at least one pointer. **A pointer is something to read, never a licence**: a label section can say "safety not established", and a study that admits children may have enrolled none.

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6. What this report cannot tell you

1. **Whether any of these drugs helps this child.** Rung 3 is one mouse paper; rung 2 is a dish. The challenge's words hold: hypotheses for follow-up.
2. **How aneuploid this child is.** The delivered data was tested with a statistic that has no power for mosaic gains; the powered analysis is planned (iteration 2, step 8) and has not run. Direction D2 inherits this gap entirely.
3. **Whether the weakened BubR1 copy is unstable or inactive.** It sits outside every catalogued domain; the structural model is confident about the region but says nothing about function. D3 is therefore argued as *more BubR1*, not *repair this BubR1*.
4. **Anything about a drug the public sources never listed** beyond what a person put in the curated table. Four of the eleven argued drugs (chloroquine, hydroxychloroquine, niacin, nicotinamide) are there only because a person read the paper; their rows say `curated_only`.
5. **The curated rungs are one person's reading of thirteen abstracts on one day.** Every row carries its PubMed id so the reading can be checked.
6. **DrugBank was not queried** (it needs a licence for programmatic access); Open Targets, ChEMBL, DGIdb, openFDA, Reactome, ClinicalTrials.gov and PubMed were.

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7. Reproducibility

Every step is one command, case-agnostic (swap the card, get another case), with a manifest:

```
uv run mva-drugs snapshot                       # card → 5 tables, 7 public APIs, cached repl
uv run python tools/track2_decoys.py ...        # a decoy card and its snapshot (×2)
uv run python tools/track2_enrich.py            # literature, paediatric trials, labels, ladde
uv run python tools/render_track2_figures.py    # the figures in this report
uv run python tools/record_track2_lineage.py    # 63 lineage edges into the warehouse ledger
```

Runtime on a laptop: the snapshot ~10 minutes cold (mostly polite waiting between requests), the enrichment ~10 minutes cold, both seconds when cached. Code: the public repository named in the submission form; every table is under `ressources/track2/`, every entry under `docs/iteration2/step-7/`. **No patient data appears anywhere in Track 2** — gene symbols, drug names, trial ids and PubMed ids only; the mechanism card names allele *classes*, never positions.

8. Where every claim lives

claim	register id	verdict · confidence	entry
D1 yields no submittable drug and becomes the do-not table	H045	supported · 0.80	113
D2 holds an approved compound with a molecule-level cell result	H046	supported · 0.70	113
D3 has an approved route on the NAD^+/SIRT2 axis with animal evidence, by proxy	H047	supported · 0.60	113
D4 holds a non-chemotherapy approved drug at rung ≥ 2 with paediatric data	H048	supported · 0.80	113
the automated stages are non-specific (a direction overlaps a decoy ≥ 0.10)	H049	supported · 0.85	112
every argued drug has a paediatric pointer	H050	supported · 0.70	113
no approved drug acts directly on BubR1	H011 (iteration 1)	supported · 0.80	078

Confidence is a transparent triage heuristic printed with its arithmetic, not a calibrated probability; the register is append-only and shows every revision, including one committed too early and corrected in the open (entry 113).